U-R-179
SEPTEMBER 15, 1954
SUPERSEDING
U-D-226
September 12, 1941

FEDERAL SPECIFICATION

RESIN, ACRYLIC, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers acrylic resin composed of any of the following types of resin: (1) polyesters of acrylic acid, (2) polyesters of substituted acrylic acids, (3) vinyl esters, (4) polystyrenes or, (5) copolymers, or mixtures, of any of the foregoing resins. The liquid for heat-curing resins shall be methyl methacrylate monomer, or methyl methacrylate monomer which has been partially polymerized.

1.2 Classification.

1.2.1 *Types, classes and colors.*—Acrylic resin covered by this specification shall be of the following types, classes and colors as specified:

Type I.—Heat-curing resins.

Class 1.—Powder and liquid.
Color A.—Pink.
Color B.—Clear.

Class 1 (a).—Liquid only.

Class 2.—Plastic cake 5 ± 3 mm. (0.20 ± 0.12 in.) thick.
Color A.—Pink.
Color B.—Clear.

Class 3.—Precured blank.
Color A.—Pink.
Color B.—Clear.

Type II.—Self-curing resins.

Class 1.—Powder and liquid.
Color A.—Pink.
Color B.—Clear.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 There are no other Federal specifications applicable to this specification except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications.—The following publication of the issue in effect on date of invitation for bids, forms a part of this specification.

American Dental Association Specification 12 for Denture Base Resin (Acrylic Resin or Mixtures of Acrylic and Other Resins).

(Copies of American Dental Association specifications may be obtained free upon application to the American Dental Association Research Fellowship, National Bureau of Standards, Washington 25, D. C.)

2.3 Specifications and other publications applicable only to individual departments are listed in section 7.

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3. REQUIREMENTS

3.1 Liquid.—The liquid shall be water clear and free of deposits or sediment.

3.2 Powder.—The powder, plastic cake, and precured blank shall be free of extraneous material such as dirt and lint.

3.3 Uncured resin, classes 1 and 2.

3.3.1 Class 1 resin when used as directed by the instructions accompanying the package (3.4.9) shall be considered ready for packing when the plasticity is such that the resin separates cleanly from the walls of the glass mixing jar (4.3.1.1). This condition shall be attained in 20 minutes or less from the start of the mixing. Five minutes later the resin when subjected to the test described in 4.3.1.2 shall be capable of being intruded into at least two holes of the die (fig. 1) to a depth of not less than 0.5 mm. (0.02 in.).

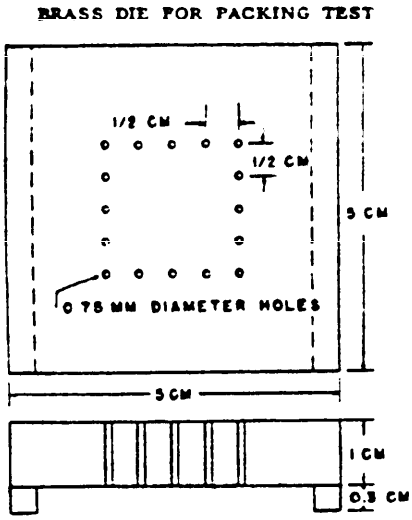


FIGURE 1.—Brass die used in testing plasticity of uncured resin.

3.3.2 Class 2 resin when treated according to the manufacturer's instructions, shall be capable of being intruded into at least two holes of the die (fig. 1) to a depth of not less than 0.5 mm. (0.02 in.).

3.4 Cured resin.

3.4.1 *Working qualities.*—The resin when processed according to instructions furnished by the manufacturer (3.4.9) shall yield a satisfactory denture.

3.4.1.1 *Glossy surface.*—The resin shall have a glossy surface when cured in a gypsum mold lined with tinfoil.

3.4.1.2 *Polish.*—The resin shall present a smooth surface having a high gloss when polished by the conventional dental methods.

3.4.2 *Color.*—The depth of color of the cured resin shall be as specified by the purchaser. The color shall be uniformly distributed or mottled. Clear resin shall be crystal clear.

3.4.3 *Translucency.*—The specimen plate (fig. 2) formed as described in 4.3.5 shall be translucent.

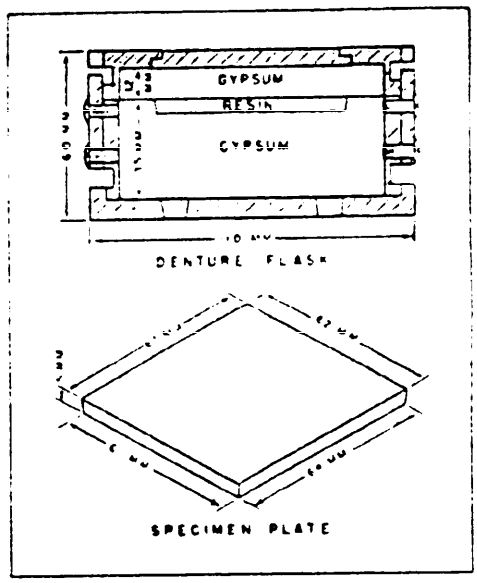


FIGURE 2.—Position of specimen plate in flask. (Buffalo Flask No. 29 to be substituted for that shown.)

3.4.4 *Porosity.*—The specimen plate (fig. 2) cut as described in 4.3.5 shall not show bubbles or voids.

3.4.5 *Sorption.*—The increase in weight of the resin shall not be more than 0.7 mg per square centimeter of surface after immersion in

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water for 24 hours at 37° ± 1°C. (98.6° ± 1.8°F.) as described in 4.3.3.

3.4.6 *Solubility.*—The loss in weight of the resin shall not be more than 0.04 mg. per square centimeter of surface when tested as described in 4.3.4.

3.4.7 *Transverse deflection.*—The deflection of at least three of the five specimens prepared as outlined in 4.3.5 shall meet the following requirements when tested in water at 37° ± 1°C. (98.6° ± 1.8°F.):

DEFLECTION LIMITS OF SPECIMENS IN TRANSVERSE TEST OF DENTURE RESINS

Load	Deflection for loads* 3,500 gm. (7.7 lb.)	1,500 gm. (3.3 lb.) to 5,000 gm. (11 lb.)	
	Maximum mm.	Minimum mm.	Maximum mm.
Type I	1.8 (0.071 in.)	2.0 (0.079 in.)	5.0 (0.197 in.)
Type II	2.5 (0.098 in.)	2.0 (0.079 in.)	6.5 (0.257 in.)

* Deflections shall be measured from an initial load of 1,500 gm. (3.3 lb.) to the load indicated.

3.4.8 *Color stability.*—The resin shall not show more than a slight change in color when tested as described in 4.3.6.

3.4.9 *Manufacturer's instructions.*—Instructions for storing and processing shall accompany each package. The instructions for type I, classes 1 and 2 resins shall include as applicable: powder-liquid ratio, temperature of flask when the resin is packed, time to bring to recommended curing temperature, time to hold at curing temperature, method of cooling flask and time to release flask. The manufacturer's instructions shall conform to those specified in 4.3.2. In addition to the instructions given in 4.3.2, the manufacturer may provide one alternate set of instructions. The manufacturer shall certify that the resin will meet the requirements of this specification when it is processed according to the alternate method as well as according to the prescribed methods. The instructions for type I, class 3 and type II resins shall be those prescribed by the manufacturer. The manufacturer's instructions shall include also a method for using the resin as a repair material.

3.5 Certification.—Acrylic resin shall be tested at the National Bureau of Standards for the purpose of establishing a certified list of material. Bidders shall attach to their bid a certificate and data showing that the brand of material they propose to furnish is the same in quality and brand as that appearing on the list of certified materials.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, sufficient resin to make not less than ten dentures (10 units) shall be taken at random from each lot delivered.

4.2 Inspection.—Visual inspection shall be used in determining compliance with the requirements outlined in 3.1, 3.2, 3.4.1.1, 3.4.1.2, 3.4.2, 3.4.3, 3.4.4, 3.4.8, 3.4.9, 5.1, and 5.2.

4.3 Physical tests.

4.3.1 *Mixing and packing characteristics.*

4.3.1.1 *Mixing.*—Class 1 resins shall be mixed as directed by the instructions accompanying the package except that the mixing temperature shall be 23° ± 1°C. (73.4° ± 1.8°F.). Likewise class 2 resins shall be brought to this temperature prior to testing. Class 1 resin shall be considered ready for packing when the plasticity is such that the resin separates cleanly from the walls of the glass mixing jar.

4.3.1.2 *Packing test.*—A 6- to 10-gm. sample of classes 1 or 2 resin shall be placed on the perforated brass die (fig. 1) to a depth of approximately 5 mm. (0.2 inch). The die and specimen shall be transferred immediately to an air chamber in which all the equipment has been preheated to within ±5°C. (±9°F.) of the packing temperature specified by the manufacturer's instructions (3.4.9). A glass plate approximately 5 mm. (0.2 in.) thick and 50 mm.

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(2.0 in.) square and a 5,000 gm. (11 pound) weight shall be placed on the specimen immediately. A sheet of cellophane shall be used as a separator between the glass plate and the specimen. The packing test for class 1 resins shall be started 5 minutes after the proper plasticity has been reached (3.3.1). Ten minutes later the die and specimen shall be removed from the air chamber and the resin stripped from the die. The depth of intrusion shall be measured with a steel scale and magnifying glass or with a dial gauge.

4.3.2 *Preparation of specimens of cured resin for sorption, solubility, transverse and color stability tests.*—Specimens of type I, classes 1 and 2 resin shall be processed in accordance with the following schedule: Ten minutes after the proper plasticity as specified in 4.3.1.2 has been reached, the flasked specimen (fig. 2), held in a flask clamp without springs, shall be submerged in water at 73° ±1°C. (163.4° ± 1.8°F.) for 8 hours or 73° ± 1°C. (163.4° ± 1.8°F.) for 1½ hours and then immersed in boiling water for ½ hour. When the heating schedule has been completed, the flask in the clamp shall be cooled in air at 23° ± 10°C. (73° ± 18°F.) for 30 minutes and then immersed in water at 23° ± 10°C. (73° ± 18°F.) for 15 minutes. For type I, class 3 and type II resins the manufacturer's instructions shall be followed.

4.3.3 *Sorption.*—The water sorption test shall be made in duplicate on disks 50 ± 1 mm. (1.97 ± 0.04 in) in diameter and 0.5 ± 0.1 mm. (0.02 ± 0.004 in.) thick (fig. 3). These specimens shall be made in a stainless steel die (fig. 3). The die shall be mounted with gypsum in the flask in the same relative position as the specimen plate in figure 2. The gypsum in the shallow half of the flask shall be poured against a glass plate to obtain a smooth surface. Trial packing shall be conducted with moist cellophane used as a separator. Before final closure of the flask a sheet of tinfoil shall be substituted for the cellophane. All surfaces of the disk shall be smooth, and the top and bottom shall be flat. The disks shall be dried in a desiccator containing anhydrous calcium sulfate ($CaSO_4$) at 37° ± 2°C. (98.6° ± 3.6°F.) for 24 hours, removed to a similar desiccator at room temperature for 1 hour, then weighed with a precision of 0.2 mg. This cycle shall be repeated until constant weight is attained, that is, until the weight loss of each disk is not more than 0.5 mg. in any 24-hour period. The disks shall then be immersed in distilled water at 37° ± 1°C. (98.6° ± 1.8°F.) for 24 hours, after which time the disks shall be removed from the water with tweezers, wiped for 30 seconds with a clean, dry hand towel, waved in the air for 15 seconds and weighed 1 minute after removal from water. The value for water sorption shall be calculated as follows for each disk:

$$\frac{\text{weight after immersion} - \text{conditioned weight (mg.)}}{\text{surface area (sq. cm.)}} = \text{sorption (mg. per sq. cm.)}$$

The average of the determined values for the two disks shall be recorded to the nearest 0.01 mg. per square centimeter. The final value (average of two determinations) shall be rounded to the nearest 0.1 mg. per square centimeter. When the final value falls midway between the two numbers, the even number shall be recorded.

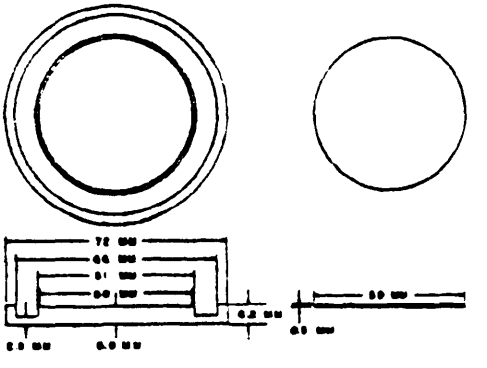


FIGURE 3.—Stainless steel mold and specimen disk prepared for use in tests for sorption, solubility and color stability.

4.3.4 *Solubility.*—After the final weighing (4.3.3), the disks shall be reconditioned to constant weight in the desiccator at 37° ± 2°C (98.6° ± 3.6°F.) as was done in 4.3.3. The soluble matter lost during immersion shall be determined to the nearest 0.01 mg. per square centimeter as follows for each disk:

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$$\frac{\text{conditioned weight (mg.) (4.3.3)} - \text{reconditioned weight (mg.) (4.3.4)}}{\text{surface area (sq. cm.)}} = \text{solubility (mg. per sq. cm.)}$$

The final value (average of two determinations) shall be rounded to the nearest 0.01 mg. per square centimeter. When the final value falls midway between two numbers, the even number shall be recorded.

4.3.5 *Transverse deflection tests.*—A plate of resin shall be processed in a gypsum mold lined with tinfoil (fig. 2). Specimens for the transverse test shall be machined from this plate to the dimensions shown in figure 4. The plate (fig. 2) shall be sawed lengthwise into five equal strips with a band saw operated at a speed of approximately 100 meters per minute (328 feet per minute). For type II resins the plate shall be at least 2 days old prior to sawing. The strips may then be sanded slightly on the molded surfaces, if necessary, to remove imperfections. All subsequent milling operations shall be with a 50 mm. (2 in.) diameter flycutter, operating at a spindle speed of approximately 1,800 revolutions per minute and at a table feed of about 15 cm. per minute (6 in. per minute). Rough cuts shall be approximately 0.08 mm. (0.003 in.) deep, and finish cuts shall be 0.03 mm. (0.001 in.) deep. The strips shall be mounted side by side in a vise and the edges machined parallel in order to remove all band saw marks. The strips at this point shall be oversize in width. One or more strips shall be mounted in a vise and the faces milled to the required thickness (2.50 ± 0.03 mm. (0.0984 ± 0.0012 in.)). All five strips shall then be clamped face to face in a vise and the edges milled to the required width (10.00 ± 0.03 mm. (0.3937 ± 0.0012 in.)). The specimens shall be stored in distilled water at 37° ± 1°C. (98.6° ± 1.8°F.) for 2 days immediately before testing. The specimen shall be mounted in an appropriate instrument, such as is shown in figure 4. An initial load, consisting of the free-moving parts of the apparatus (fig. 4) plus sufficient lead shot to make a load of 1,500 gm. (3.3 pounds), shall be placed on the specimen. Each 500 gm. increments of shot shall be added at a uniform rate during the last 30 seconds of each minute. The following schedule shall be used in applying the loads and in making the observations:

SCHEDULE OF LOADING AND READING

Time		Load	Dial reading*
min.	sec.	gm.	mm.
0	00	1,500	
0	30	1,500	A
1	00	2,000	
1	30	2,000	
2	00	2,500	
2	30	2,500	
3	00	3,000	
3	30	3,000	
4	00	3,500	
4	30	3,500	B
5	00	4,000	
5	30	4,000	
6	00	4,500	
6	30	4,500	
7	00	5,000	
7	30	5,000	C

Deflection at 3,500 gm. = B – A
Deflection at 5,000 gm. = C – A
* Readings shall be made to the nearest 0.01 mm.

The difference between the deflections at the initial load and at specified loads shall be taken as the deflections of the specimen. The recorded deflections shall be reported to the nearest 0.1 mm. When the determined number falls midway between two numbers, the even number shall be recorded.

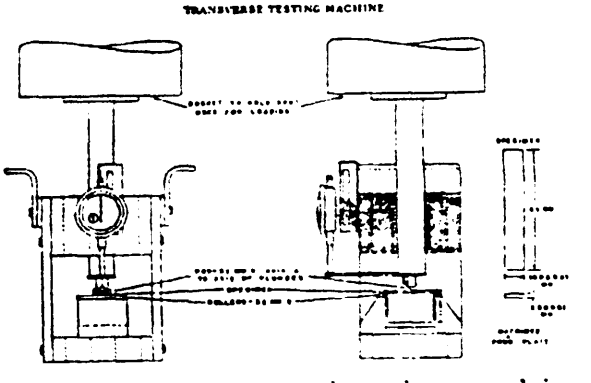


FIGURE 4.—Apparatus and specimen used in transverse test. Lower section of apparatus is submerged in water at 37°C. (98.6°F.) to cover specimen.

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4.3.6 *Color stability.*—One of the disks used in 4.3.3 and 4.3.4 shall be exposed to the radiation of a lamp (fig. 5) with an S–1 bulb which has been in use not less than 50 nor more than 400 hours. The light source shall be a combination tungsten filament mercury-arc enclosed in Corex D or other glass which has a low transmission below 2,800 A. The lamp shall be rated at 400 watts. The specimen shall be carried on an aluminum disk which shall be centered under the S–1 bulb and which shall rest on a phonograph turntable, operated at 33 revolutions per minute. The specimen shall be held by screws approximately 5 mm. (0.2 in.) above the disk and 12.7 cm. (5 in.) from the center and shall be maintained between 60 and 65° C. (140 and 149° F.). The plane of the top surface of the specimen shall be 17.8 cm. (7 in.) from the bottom of the S–1 bulb. After exposure for 24 hours, the specimen, compared with an unexposed duplicate specimen, shall not show more than a slight change in color (perceptible with difficulty). The comparison shall be made by visual inspection in daylight.

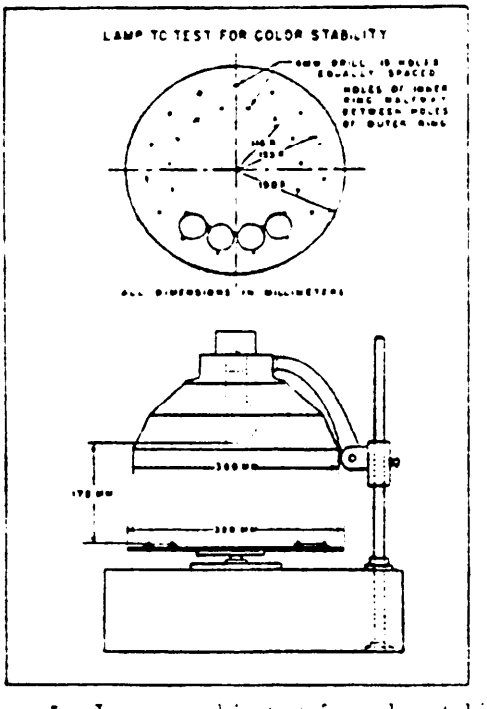


FIGURE 5.—Lamp used in test for color stability of resin.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, the material shall be supplied in properly sealed glass, polyethylene or metal containers, or in aluminum envelopes for packages which contain more than 100 units (2, 100 gm.). For packages which contain more than 100 units (2, 100 gm.) the containers shall be made of materials which shall not contaminate or permit contamination of the resin.

5.2 Packing.—Unless otherwise specified, acrylic resin shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery.

5.3 Marking.

5.3.1 *Lot numbers.*—Each container of material shall be marked with a serial number or a combination of letters and numbers which shall refer to the manufacturer's records for that particular lot or batch.

5.3.2 *Date of manufacture.*—The date of manufacture (year and month) shall be given on the container either as a separate item or as a part of the lot number (5.3.1).

5.3.3 *Net weight and volume.*—The minimum net weight in grams of powder and the minimum net volume in milliliters of liquid shall be given in legible type on the container. When the term "unit" is used as a designation, it shall represent not less than 21 grams of powder and a sufficient volume of liquid to combine with the powder.

6. NOTES

6.1 Intended use.—The acrylic resin covered by this specification is intended for use in the molding of the dentures. If an inhibitor is recommended by the manufacturer, it shall be supplied with the unit package.

6.2 Purchasers should specify type, class and color required and should exercise any desired options offered herein. (See 1.2, 4.1, 5.1, 5.2 and 7.2.3.)

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6.3 Certified materials.—Details of the procedure for establishing lists of certified materials are given in the Journal of the American Dental Association, Vol. 45, pages 585-590 (Nov. 1952).

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications cover only the types, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and are not intended to include all of the types, etc., which are commercially available.

6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–551–7050—Resin, Acrylic, Dental, Clear.
6520–551–7550—Resin, Acrylic, Dental Pink.
6520–551–8150—Resin, Acrylic, Dental, Denture Base Repair, 1½ Lb.
6520–551–8550—Acrylic Resin Liquid, Dental, 8 oz.

6.7 Transportation description.—Transportation description applicable to this item is:

Dental plaster, suitable for making dental casts or molds.
Carload weight 40,000 pounds.
Truckload weight 40,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
PPP–B–676—Boxes, Set-Up, Paperboard.

Military Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes: Wood, Nailed.
MIL–L–10547—Liners, Case, Waterproof.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *Unit of issue.*

7.2.2.1.1 *Type I, class 1.*—One container of powder having a minimum net weight of 210 grams, one container of liquid containing 10 percent in excess of the quantity required to combine with the powder, and one container of inhibitor when supplied (see 6.1) sufficient to process the powder, constitute one unit.

7.2.2.1.2 *Type I, class 1 (a).*—Eight (8) ounces of liquid, as specified, constitute one unit.

7.2.2.1.3 *Type II, class 1.*—One-half-pound container of powder and a quantity of liquid 10 percent in excess of the quantity required to combine with the powder, constitute one unit.

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7.2.2.2 *Immediate containers.*—Powders shall be packaged in wide-mouth glass bottles of suitable size and design. Liquids shall be packaged in narrow-mouth, light-resistant glass bottles of suitable size and design.

7.2.2.2.1 *Closure.*—Closure shall be a metal continuous-thread screwcap, treated inside and outside to resist corrosion. Cap shall have a nonabsorbent lining material which must remain in cap when cap is removed from bottle. Lining material shall be resilient in itself, or shall be backed by resilient material. Cap shall be tightened to prevent leakage. Bottle shall be further sealed by an inner seal which shall cover and adhere to mouth of bottle, capable of preventing loss of contents should the cap become loosened, or by a "Gel" or similar viscose cap covering the stopper externally and extending well down on the neck of bottle. When "Gel" or similar viscose cap coverings are used, they shall be treated to inhibit the growth of fungus or mildew.

7.2.2.2.2 *Compatibility of container.*—The container, closure, and lining shall not interact physically or chemically with the substance contained so as to be altered or to alter the strength, quality, or purity of the container contents.

7.2.2.3 *Unit packaging.*

7.2.2.3.1 *Type I, class 1 (a).*—Unit packaging is not required.

7.2.2.3.2 *Type I, class 1 and type II, class 1.*—Each unit of issue shall be packaged in a carton constructed in accordance with Federal Specification LLL–B–631 or PPP–B–676. Bottles shall be protected by flexible corrugated fiberboard sleeves or by suitable interior fittings in such a manner as to secure the bottles in a vertical position and to insure against shifting or damage. Each carton shall be adequately secured to prevent accidental opening.

7.2.2.4 *Intermediate packaging—type I, class 1 (a) only.*—Twelve bottles (12 units) shall be packaged in a carton fabricated of not less than 200-pound test fiberboard and constructed in accordance with Federal Specification LLL–B–631. Carton shall be provided with full-height partitions fabricated of not less than 200-pound test fiberboard. Carton shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—The number of unit or intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load. Construction, closure, and strapping (when required) shall be in accordance with the requirements of Federal Specification NN–B–621. Arrangement shall be such that bottles are packed in a vertical position.

7.2.3.2 *For overseas shipment.*—The number of units or intermediate cartons specified in table I shall be overpacked in an exterior container designed for type 3 load and constructed in accordance with the requirements of Military Specification JAN–P–106. Arrangement shall be such that bottles are packed in a vertical position.

7.2.3.2.1 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to

TABLE I.– *Packing quantities*

Stock Nos.	Type	Class	Number of units or intermediate cartons to be overpacked in each shipping container	Total number of units to be contained in each shipping container
6520–551–7050	I	1	24	24
6520–551–7550	I	1	24	24
6520–551–8150	II	1	24	24
6520–551–8550	I	1 (a)	4	48

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the requirements of Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.3.2.2 *Strapping.*—Strapping, when required, shall be in accordance with applicable box specification.

NOTE.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution

7.2.3.3 *Packing variation permitted.*—If the number of units in any shipment is less than the specified number of units to be overpacked in an exterior container, such units shall be overpacked in an exterior container of suitable size and design which will insure safe delivery to destination.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.*—Each bottle shall be commercially identified and shall show the lot (control) number. Each bottle of liquid shall also be marked with the word *"Inflammable"*. Each unit carton and each intermediate carton shall be marked with stock number, nomenclature, type (i. e., powder and liquid), lot (control) number, office identifying number, type of resin (i. e., if the powder is a mixture of methyl and ethyl methacrylate it shall be stated), date of manufacture (month, year), unit, quantity, name of contractor, and the legend: "Store in a cool, dark place." Cartons of type I material shall also be marked with the quantity of powder in *denture units*.

7.2.4.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Ordering data.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.1 and 7.2.3.2.)

MILITARY INTEREST:

Army—CM
Navy—Md
Air Force—USAF.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines of reverse side, staple or cement, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 62 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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with the powder and one container of